

Chemistry

SECTION 1 (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

- *Q.1 According to Bohr's model, the highest kinetic energy is associated with the electron in the
- (A) first orbit of H atom (B) first orbit of He^+
 (C) second orbit of He^+ (D) second orbit of Li^{2+}

Ans. **B**

Sol. $\text{K.E.} = +13.6 \frac{Z^2}{n^2} \text{ eV}$

- (A) $n = 1, z = 1$
 (B) $n = 1, z = 2$
 (C) $n = 2, z = 2$
 (D) $n = 2, z = 3$

- K. E. = +13.6 eV
 K. E. = +54.4 eV
 K. E. = +13.6 eV
 K. E. = +30.6 eV

- Q.2 In a metal deficient oxide sample, $\text{M}_x\text{Y}_2\text{O}_4$ (M and Y are metals), M is present in both +2 and +3 oxidation states and Y is in +3 oxidation state. If the fraction of M^{2+} ions present in M is $\frac{1}{3}$, the value of X is ____.
- (A) 0.25 (B) 0.33
 (C) 0.67 (D) 0.75

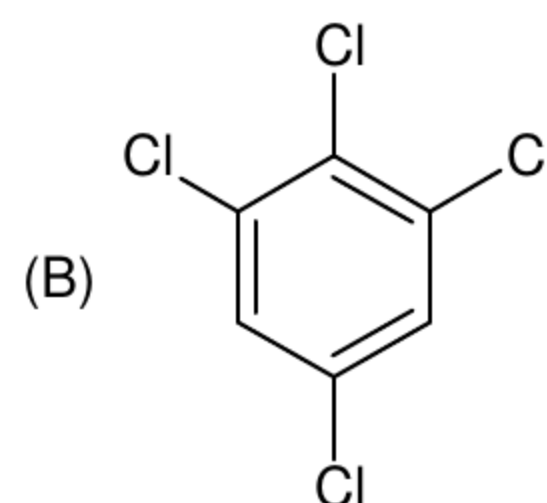
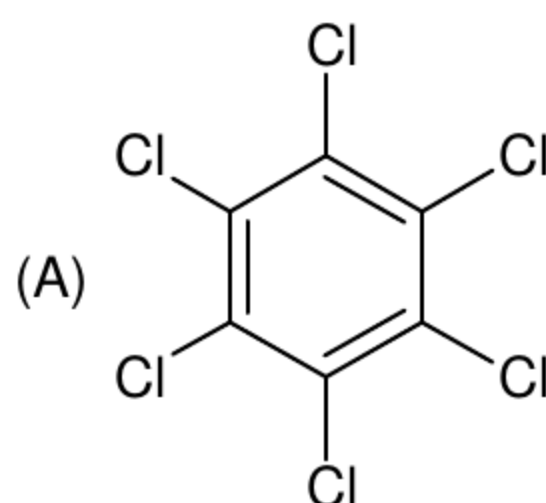
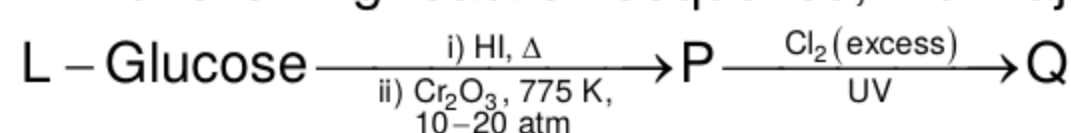
Ans. **D**

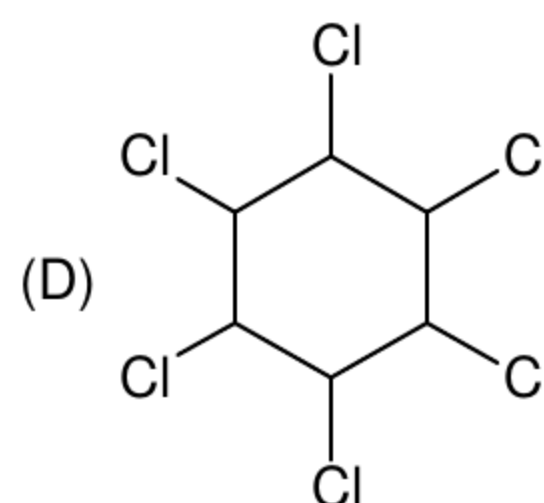
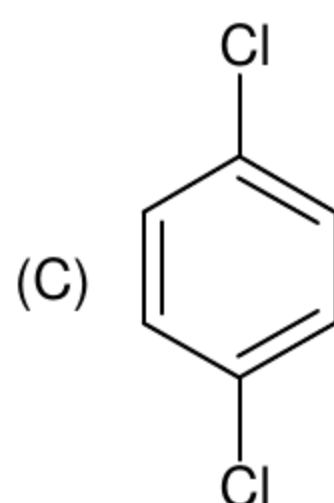
Sol. $\text{M}_x\text{Y}_2\text{O}_4$

$$+2 \times \frac{x}{3} + 3 \times \frac{2x}{3} = +2$$

$$x = 0.75$$

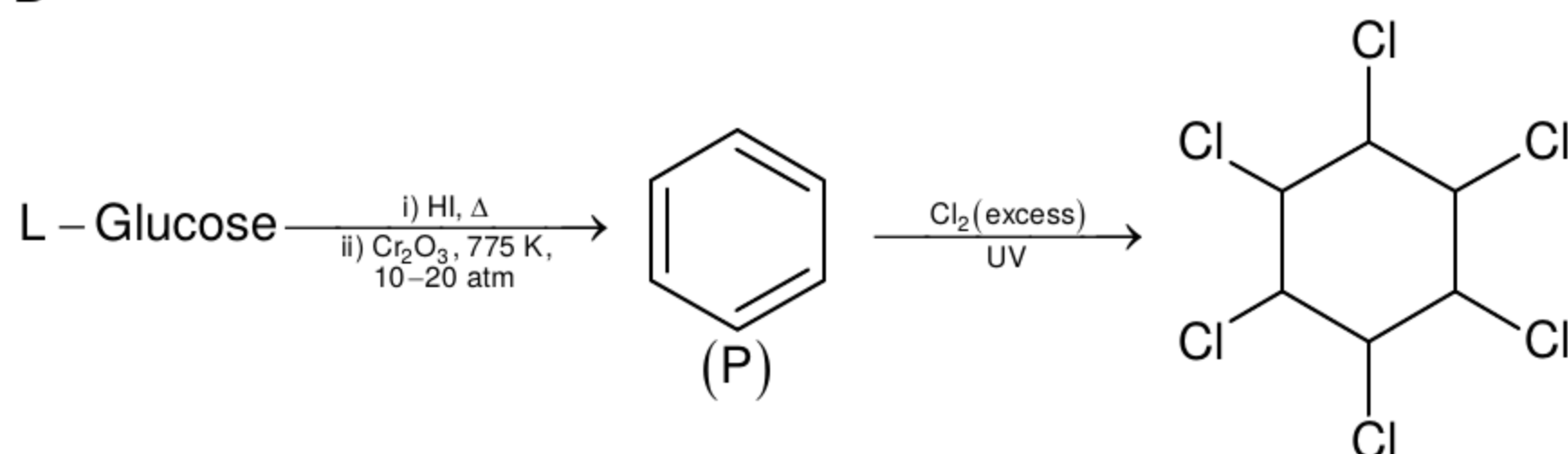
- Q. 3 In the following reaction sequence, the major product Q is





Ans. D

Sol.

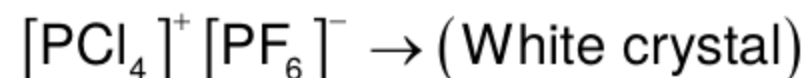
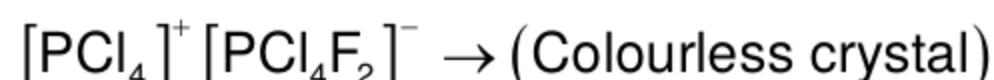


Q.4 The species formed on fluorination of phosphorus pentachloride in a polar organic solvent are

- (A) $[\text{PF}_4]^+ [\text{PF}_6]^-$ and $[\text{PCl}_4]^+ [\text{PF}_6]^-$ (B) $[\text{PCl}_4]^+ [\text{PCl}_4\text{F}_2]^-$ and $[\text{PCl}_4]^+ [\text{PF}_6]^-$
 (C) PF_3 and PCl_3 (D) PF_5 and PCl_3

Ans. B

Sol. On fluorination of PCl_5 in polar organic solvent ionic isomers are formed, i.e.



SECTION 2 (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
Zero Marks : 0 If unanswered;
- Negative Marks* : -2 In all other cases. · For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then
 choosing **ONLY** (A), (B) and (D) will get +4 marks;
 choosing **ONLY** (A) and (B) will get +2 marks;
 choosing **ONLY** (A) and (D) will get +2 marks;
 choosing **ONLY** (B) and (D) will get +2 marks;
 choosing **ONLY** (A) will get +1 mark;
 choosing **ONLY** (B) will get +1 mark;
 choosing **ONLY** (D) will get +1 mark;
 choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
 choosing any other option(s) will get -2 marks.

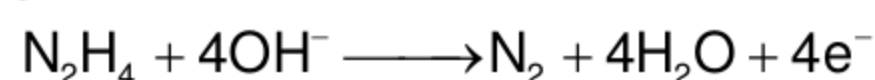
- Q. 5 An aqueous solution of hydrazine (N_2H_4) is electrochemically oxidized by O_2 , thereby releasing chemical energy in the form of electrical energy. One of the products generated from the electrochemical reaction is $\text{N}_2(\text{g})$.

Choose the correct statement(s) about the above process

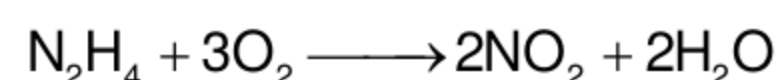
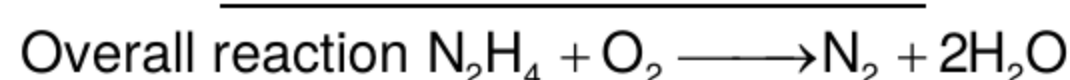
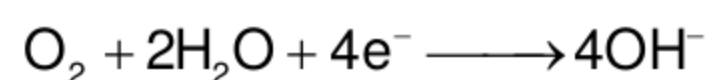
- (A) OH^- ions react with N_2H_4 at the anode to form $\text{N}_2(\text{g})$ and water, releasing 4 electrons to the anode.
 (B) At the cathode, N_2H_4 breaks to $\text{N}_2(\text{g})$ and nascent hydrogen released at the electrode reacts with oxygen to form water.
 (C) At the cathode, molecular oxygen gets converted to OH^- .
 (D) Oxides of nitrogen are major by-products of the electrochemical process.

Ans. **A, C, D**

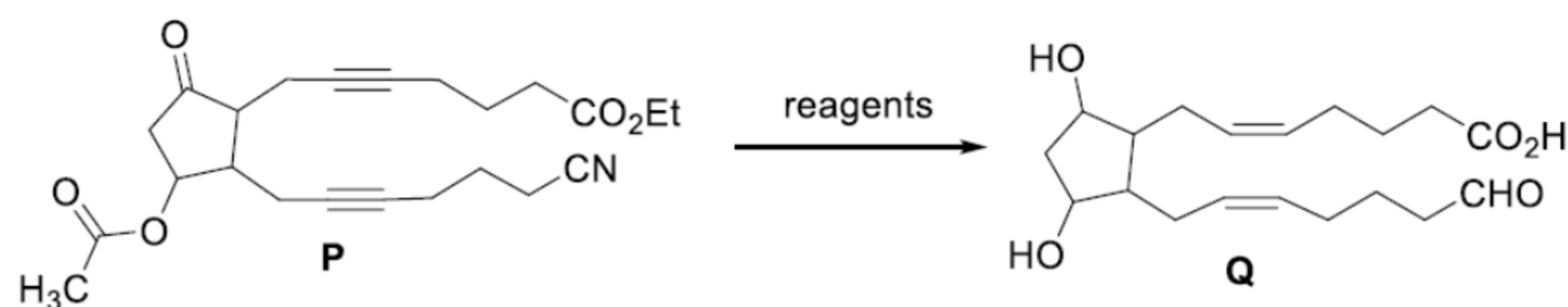
Sol. At anode



At cathode



- Q. 6 The option(s) with correct sequence of reagents for the conversion of **P** to **Q** is(are)



- (A) i) Lindlar's catalyst, H_2 ; ii) SnCl_2/HCl ; iii) NaBH_4 ; iv) H_3O^+
 (B) i) Lindlar's catalyst, H_2 ; ii) H_3O^+ ; iii) SnCl_2/HCl ; iv) NaBH_4
 (C) i) NaBH_4 ; ii) SnCl_2/HCl ; iii) H_3O^+ ; iv) Lindlar's catalyst, H_2 ;
 (D) i) Lindlar's catalyst, H_2 ; ii) NaBH_4 ; iii) SnCl_2/HCl ; iv) H_3O^+

Ans. **C, D**

Sol. Reagent

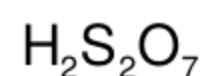
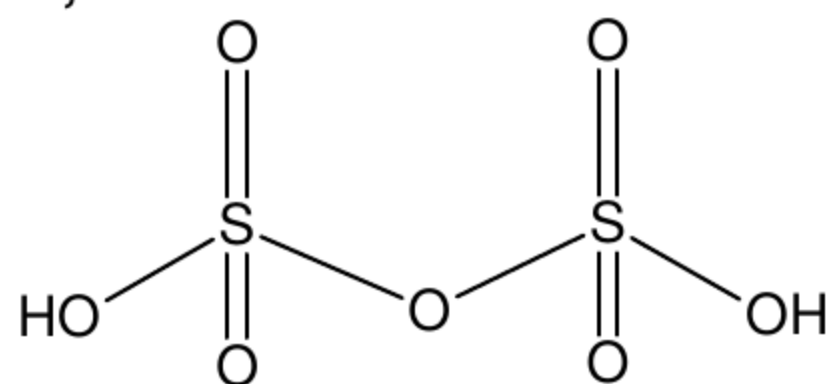
- (C) i) NaBH_4 ; ii) SnCl_2/HCl ; iii) H_3O^+ ; iv) Lindlar's catalyst, H_2 ;
 (D) i) Lindlar's catalyst, H_2 ; ii) NaBH_4 ; iii) SnCl_2/HCl ; iv) H_3O^+

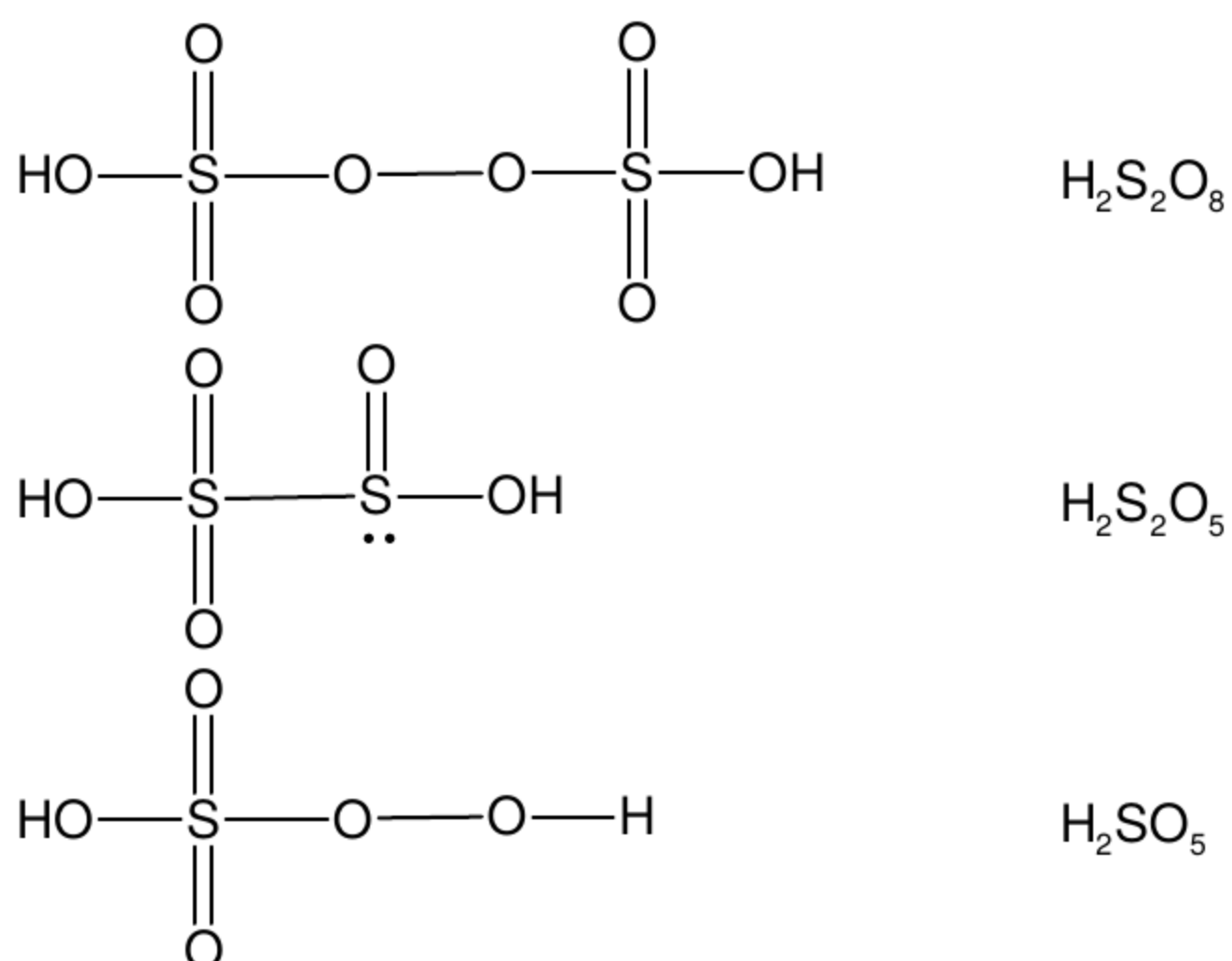
- *Q. 7 The compound(s) having peroxide linkage is(are)

- (A) $\text{H}_2\text{S}_2\text{O}_7$ (B) $\text{H}_2\text{S}_2\text{O}_8$
 (C) $\text{H}_2\text{S}_2\text{O}_5$ (D) H_2SO_5

Ans. **B, D**

Sol.



**SECTION 3 (Maximum Marks: 24)**

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct integer is entered;
Zero Marks : 0 In all other cases.

- Q. 8 To form a complete monolayer of acetic acid on 1g of charcoal, 100 mL of 0.5 M acetic acid was used. Some of the acetic acid remained unadsorbed. To neutralize the unadsorbed acetic acid, 40 mL of 1 M NaOH solution was required. If each molecule of acetic acid occupies $\mathbf{P} \times 10^{-23} \text{ m}^2$ surface area on charcoal, the value of **P** is _____.
 [Use given data : Surface area of charcoal = $1.5 \times 10^2 \text{ m}^2 \text{ g}^{-1}$; Avogadro's number (N_A) = $6.0 \times 10^{23} \text{ mol}^{-1}$]

Ans. 2500

Sol. Number of moles of NaOH = $40 \times 1 \times 10^{-3} = 4 \times 10^{-2}$
 Number of moles of Acetic acid = $100 \times 0.5 \times 10^{-3} = 5 \times 10^{-2}$
 $\therefore$ Number of moles of acetic acid adsorbed on charcoal = $5 \times 10^{-2} - 4 \times 10^{-2} = 1 \times 10^{-2}$
 $\therefore$ Number of molecules of acetic acid adsorbed = $1 \times 10^{-2} \times N_A$
 $= 6 \times 10^{21}$
 Total surface area adsorbed by one molecule of acetic acid = $\frac{1.5 \times 10^2}{6 \times 10^{21}} = 2500 \times 10^{-23}$
 $\therefore \mathbf{P} = 2500$

- Q. 9 Vessel-1 contains $\mathbf{w}_2$ g of a non-volatile solute **X** dissolved in $\mathbf{w}_1$ g of water. Vessel-2 contains $\mathbf{w}_2$ g of another non-volatile solute **Y** dissolved in $\mathbf{w}_1$ g of water. Both the vessels are at the same temperature and pressure. The molar mass of **X** is 80% of that of **Y**. The van't Hoff factor for **X** is 1.2 times of that of **Y** for their respective concentrations. The elevation of boiling point for solution in Vessel-1 is _____% of the solution in Vessel-2.

Ans. 150

Sol. $(\Delta T_b)_1 = i_1 k_f \times \frac{W_2}{\frac{M_x}{\left(\frac{W_1}{1000}\right)}}$

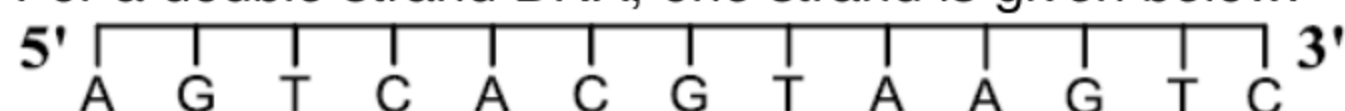
$(\Delta T_b)_2 = i_2 k_f \times \frac{W_2}{\frac{M_y}{\left(\frac{W_1}{1000}\right)}}$

$\frac{(\Delta T_b)_1}{(\Delta T_b)_2} = \frac{i_1}{i_2} \times \frac{M_y}{M_x} = 1.2 \times \frac{1}{0.8} = 1.5$

$(\Delta T_b)_1 = 1.5(\Delta T_b)_2$

Ans. 150

Q. 10 For a double strand DNA, one strand is given below:



The amount of energy required to split the double strand DNA into two single strands is _____ kcal mol⁻¹.

[Given: Average energy per H-bond for A-T base pair = 1.0 kcal mol⁻¹, G-C base pair = 1.5 kcal mol⁻¹, and A-U base pair = 1.25 kcal mol⁻¹. Ignore electrostatic repulsion between the phosphate groups.]

Ans. 41

Sol. Seven A, T and Six G, C pairs are there.
Energy required = $7 \times 2 \times 1 + 6 \times 3 \times 1.5 = 41$ kcal

*Q. 11 A sample initially contains only U-238 isotope of uranium. With time, some of the U-238 radioactively decays into Pb-206 while the rest of it remains undisintegrated.

When the age of the sample is $P \times 10^8$ years, the ratio of mass of Pb-206 to that of U-238 in the sample is found to be 7. The value of P is _____.

[Given: Half-life of U-238 is 4.5×10^9 years; $\log_e 2 = 0.693$]

Ans. 142.65 or 143

Sol. Mass = Number of moles $\times$ Molar mass

$$\frac{(\text{Mass})_{\text{Pb}}}{(\text{Mass})_{\text{U}}} = \frac{n_{\text{Pb}} \times 206}{n_{\text{U}} \times 238} = \frac{7}{1}$$

$$\frac{n_{\text{Pb}}}{n_{\text{U}}} = \frac{238 \times 7}{206}$$

$$\ln\left(1 + \frac{n_{\text{Pb}}}{n_{\text{U}}}\right) = \lambda t$$

$$\ln\left(1 + \frac{238 \times 7}{206}\right) = \frac{\ln 2}{t_{1/2}} \times t$$

$$\ln\left(\frac{206 + 1666}{206}\right) = \frac{\ln 2}{t_{1/2}} \times t$$

$$\ln(9) = \frac{\ln 2}{t_{1/2}} \times t$$

$$t = \frac{\ln 9}{\ln 2} \times t_{1/2}$$

$$t = \frac{2 \times 0.4771}{0.3010} \times 4.5 \times 10^9$$

$$t = 14.265 \times 10^9 = 142.65 \times 10^8 \text{ years}$$

$$P = 142.65$$

- Q. 12 Among $[\text{Co}(\text{CN})_4]^{4-}$, $[\text{Co}(\text{CO})_3(\text{NO})]$, XeF_4 , $[\text{PCl}_4]^+$, $[\text{PdCl}_4]^{2-}$, $[\text{ICl}_4]^-$, $[\text{Cu}(\text{CN})_4]^{3-}$ and P_4 the total number of species with tetrahedral geometry is_____.

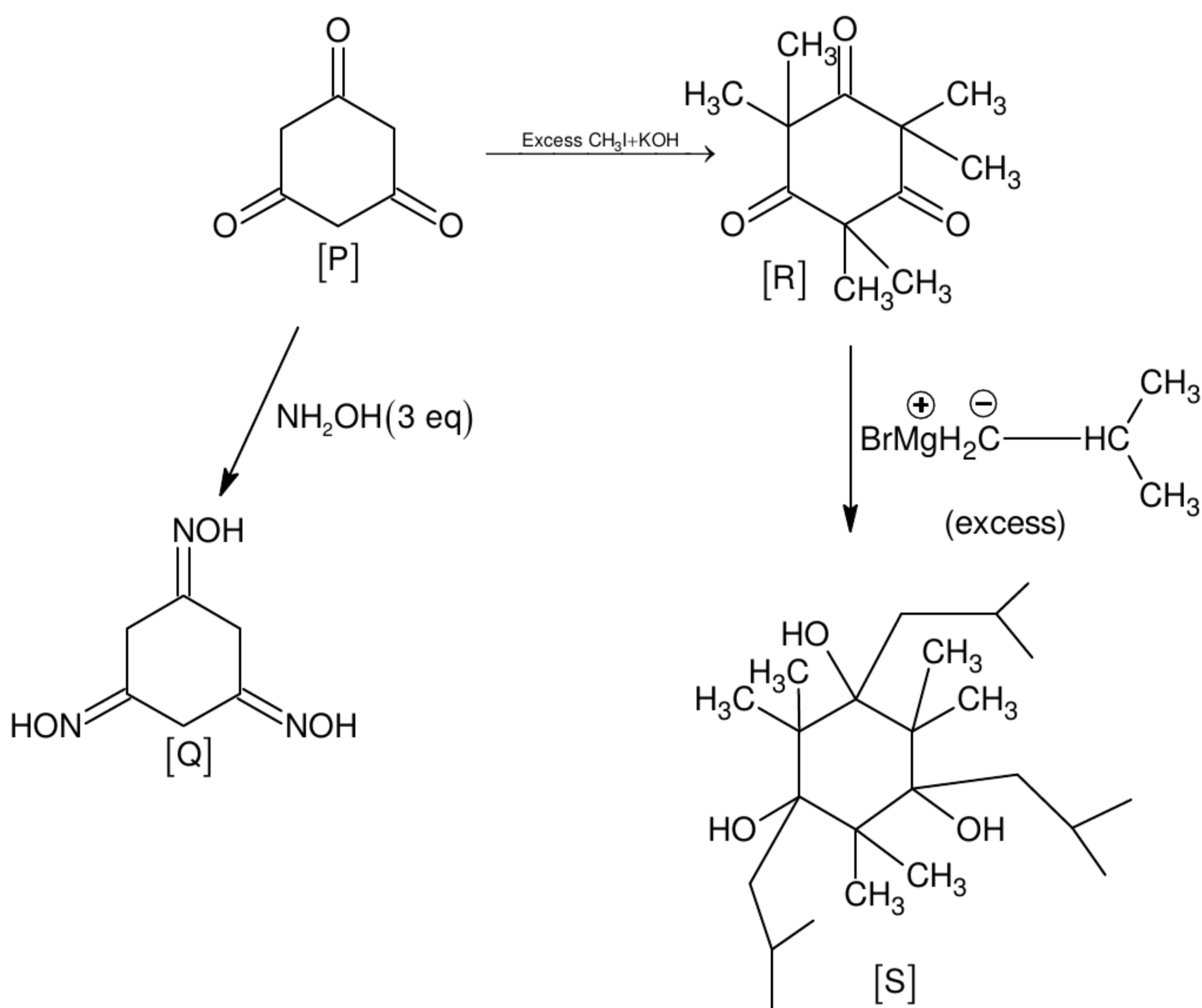
Ans. 5

Sol. $[\text{Co}(\text{CN})_4]^{4-} \rightarrow \text{Tetrahedral}$
 $[\text{Co}(\text{CO})_3(\text{NO})] \rightarrow \text{Tetrahedral}$
 $\text{XeF}_4 \rightarrow \text{Square planar}$
 $[\text{PCl}_4]^+ \rightarrow \text{Tetrahedral}$
 $[\text{PdCl}_4]^{2-} \rightarrow \text{Square planar}$
 $[\text{ICl}_4]^- \rightarrow \text{Square planar}$
 $[\text{Cu}(\text{CN})_4]^{3-} \rightarrow \text{Tetrahedral}$
 $\text{P}_4 \rightarrow \text{Tetrahedral}$

- Q. 13 An organic compound **P** having molecular formula $\text{C}_6\text{H}_6\text{O}_3$ gives ferric chloride test and does not have intramolecular hydrogen bond. The compound **P** reacts with 3 equivalents of NH_2OH to produce oxime **Q**. Treatment of **P** with excess methyl iodide in the presence of KOH produces compound **R** as the major product. Reaction of **R** with excess *iso*-butylmagnesium bromide followed by treatment with H_3O^+ gives compound **S** as the major product.
 The total number of methyl ($-\text{CH}_3$) group(s) in compound **S** is_____.

Ans. 12

Sol.



$\therefore$ Number of methyl group in product (S) = 12

SECTION 4 (Maximum Marks: 12)

- This section contains **TWO (02)** paragraphs.
- Based on each paragraph, there are **TWO (02)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered in the designated place;
Zero Marks : 0 In all other cases.

“PARAGRAPH I”

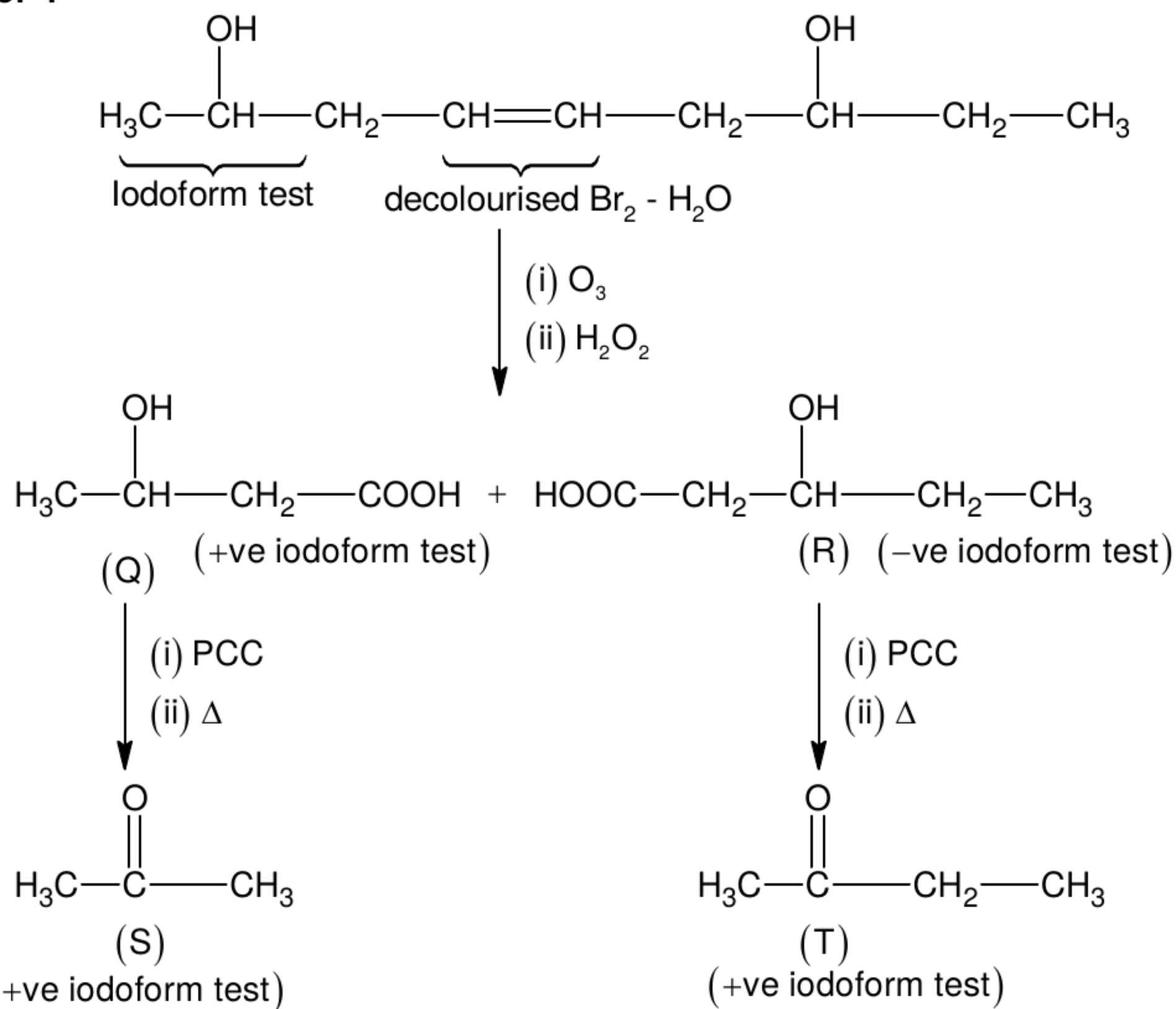
An organic compound **P** with molecular formula $\text{C}_9\text{H}_{18}\text{O}_2$ decolorizes bromine water and also shows positive iodoform test. **P** on ozonolysis followed by treatment with H_2O_2 gives **Q** and **R**. While compound **Q** shows positive iodoform test, compound **R** does not give positive iodoform test. **Q** and **R** on oxidation with pyridinium chlorochromate (PCC) followed by heating give **S** and **T**, respectively. Both **S** and **T** show positive iodoform test.

Complete copolymerization of 500 moles of **Q** and 500 moles of **R** gives one mole of a single acyclic copolymer **U**.

[Given, atomic mass: H = 1, C = 12, O = 16]

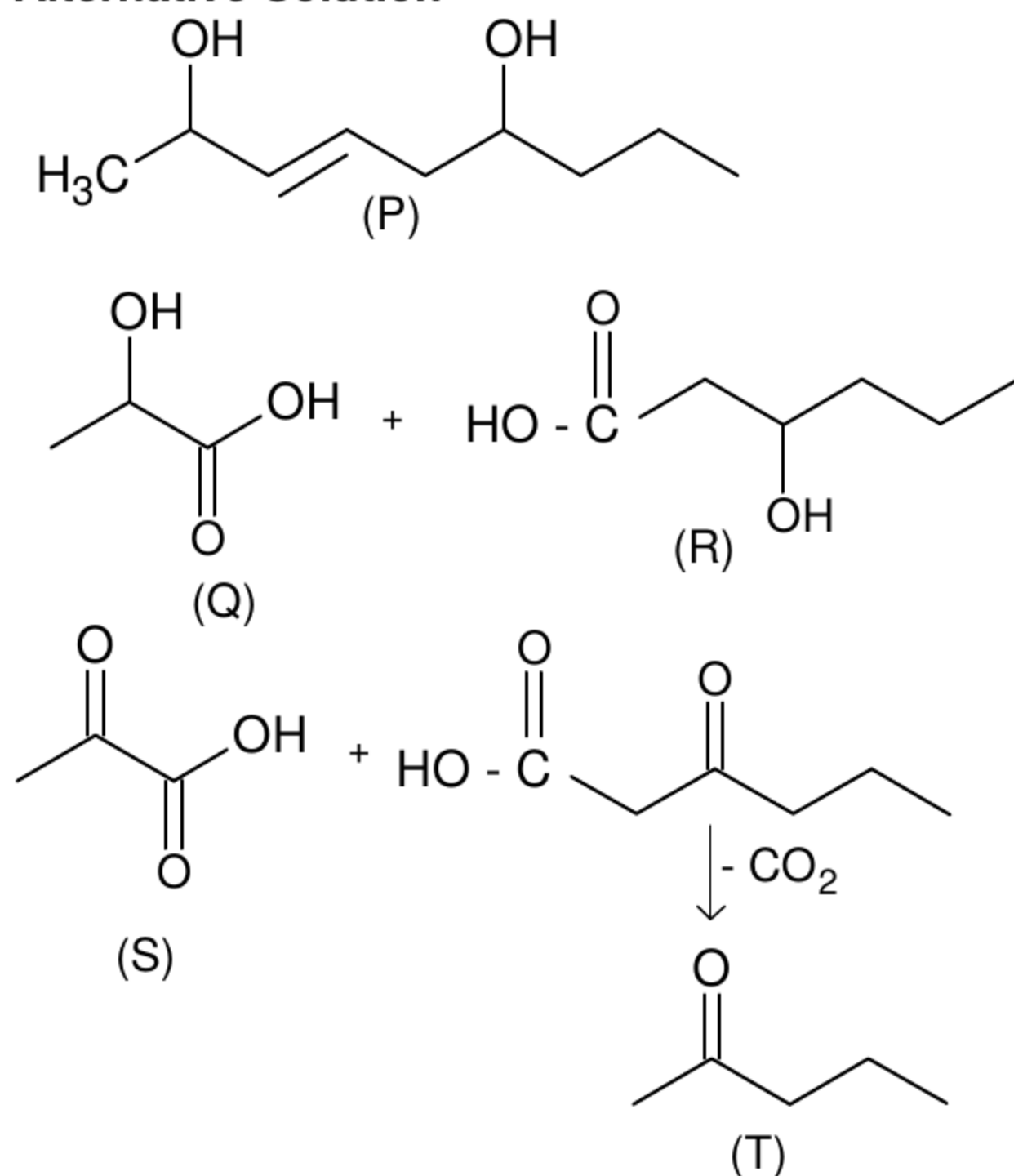
Q.14 Sum of number of oxygen atoms in **S** and **T** is _____.

Ans. 2 or 4
Sol.



Sum of oxygen atom in S and T = 1 + 1 = 2

Alternative Solution



Sum of O atom = 3 + 1 = 4

"PARAGRAPH I"

An organic compound **P** with molecular formula $C_9H_{18}O_2$ decolorizes bromine water and also shows positive iodoform test. **P** on ozonolysis followed by treatment with H_2O_2 gives **Q** and **R**. While compound **Q** shows positive iodoform test, compound **R** does not give positive iodoform test. **Q** and **R** on oxidation with pyridinium chlorochromate (PCC) followed by heating give **S** and **T**, respectively. Both **S** and **T** show positive iodoform test.

Complete copolymerization of 500 moles of **Q** and 500 moles of **R** gives one mole of a single acyclic copolymer **U**.

[Given, atomic mass: H = 1, C = 12, O = 16]

Q.15 The molecular weight of **U** is _____.

Ans. 93018

Sol. $Q + R \longrightarrow \text{Polymer (U)}$

Since moles of **U** = 1.

$$\text{Moles} = \frac{\text{Weight}}{\text{Molar mass}}$$

Weight of '**U**' = Weight of (Q + R) – Weight of water

Weight of '**Q**' = $500 \times 104 = 52,000 \text{ g}$

Weight of '**R**' = $500 \times 118 = 59,000 \text{ g}$

Weight of ' H_2O ' = $999 \times 18 = 17982 \text{ g}$

Weight of polymer = $52000 + 59000 - 17982$
= 93018 g/mol

"PARAGRAPH II"

When potassium iodide is added to an aqueous solution of potassium ferricyanide, a reversible reaction is observed in which a complex **P** is formed. In a strong acidic medium, the equilibrium shifts completely towards **P**. Addition of zinc chloride to **P** in a slightly acidic medium results in a sparingly soluble complex **Q**.

Q.16 The number of moles of potassium iodide required to produce two moles of **P** is _____.

Ans. 3

Sol. $3KI + 2K_3[Fe(CN)_6] \longrightarrow 2K_4[Fe(CN)_6] + KI_3$
(P)

"PARAGRAPH II"

When potassium iodide is added to an aqueous solution of potassium ferricyanide, a reversible reaction is observed in which a complex **P** is formed. In a strong acidic medium, the equilibrium shifts completely towards **P**. Addition of zinc chloride to **P** in a slightly acidic medium results in a sparingly soluble complex **Q**.

Q.17 The number of zinc ions present in the molecular formula of **Q** is _____.

Ans. 2 or 3

Sol. $3ZnCl_2 + 2K_4[Fe(CN)_6] \longrightarrow K_2Zn_3[Fe(CN)_6]_2$
Excess White ppt. (Q)

$2ZnCl_2 + K_4[Fe(CN)_6] \longrightarrow Zn_2[Fe(CN)_6] + 4KCl$
White ppt. (Q)